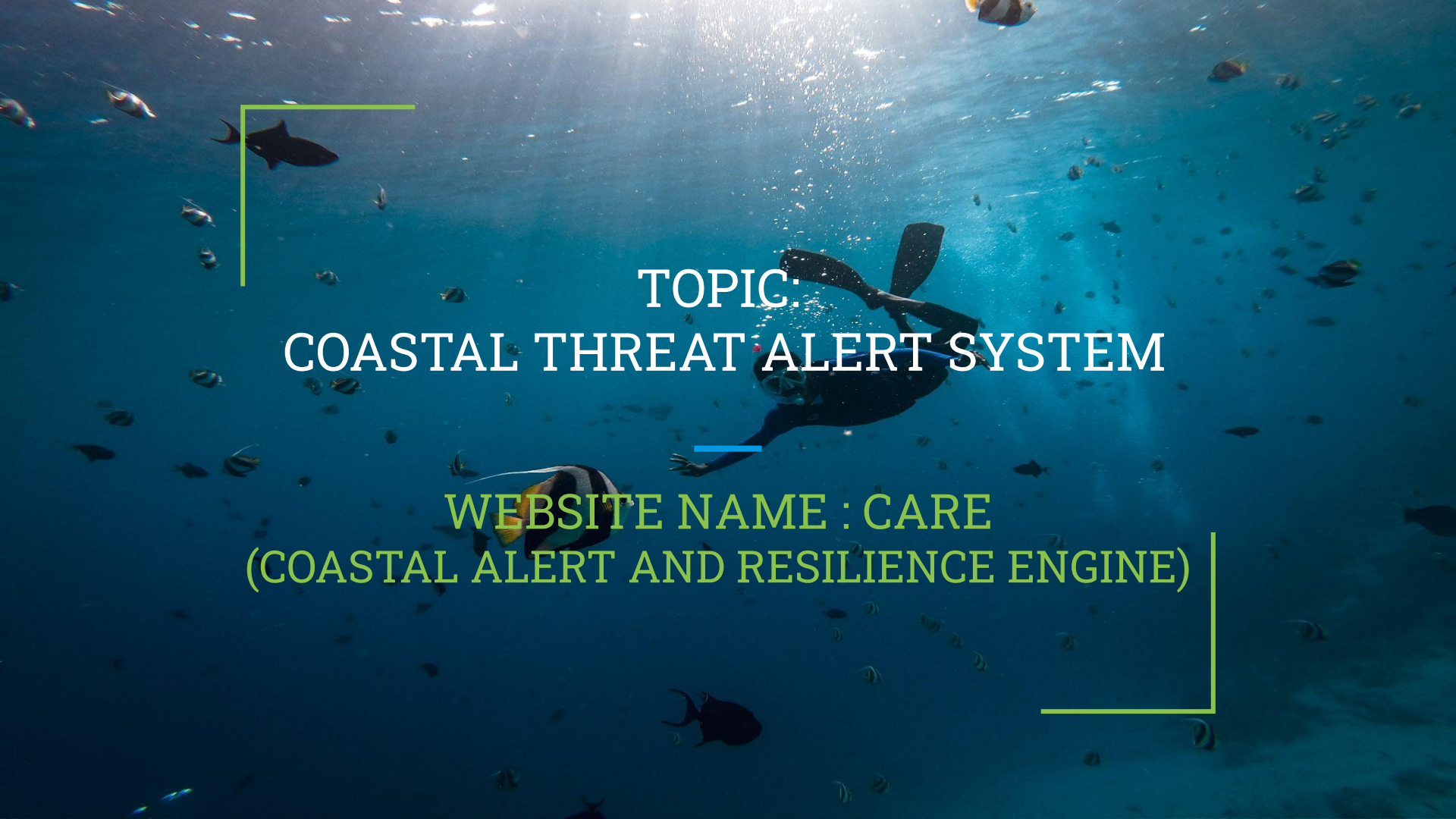


TOPIC:
COASTAL THREAT ALERT SYSTEM

WEBSITE NAME : CARE
(COASTAL ALERT AND RESILIENCE ENGINE)

PROBLEM STATEMENT

- Coastal regions face threats: cyclones, rising sea levels, pollution, erosion
- Lack of a unified real-time monitoring & alert system
- Need for early warning solutions to protect lives, habitats, and infrastructure



PROPOSED SOLUTION

- AI-powered platform for early detection of coastal threats
- Data from sensors, satellites, and historical records
- Predict floods, algal blooms, cyclones, etc.
- Real-time alerts via SMS, app notifications, and dashboards

KEY FEATURE

- Data integration from weather stations & satellites
- AI/ML predictive analytics
- Multi-channel alert system
- Scalable and community-focused solution

TECH STACK

- Frontend: React.js (Dashboards)
- Backend: Python/Node.js
- AI/ML: TensorFlow / PyTorch
- Data: OpenWeather API, ISRO Satellite Data
- Notifications: Firebase, Twilio
- Database: PostgreSQL / MongoDB
- Hosting: AWS / GCP

SYSTEM ARCHITECTURE

Data Sources → AI/ML Analysis → Alert Engine → Web/Mobile Dashboards → Users

IMPACT

- Protects lives via early alerts
- Minimizes economic & infrastructure Damage.
- Preserves blue carbon habitats
- Strengthens disaster preparedness

IMPLEMENTATION PLANNING

- Phase 1: Data Integration → Day 1
- Phase 2: AI Model Development → Day 1
- Phase 3: Dashboard & App → Day 2
- Phase 4: Testing → Day 2
- Phase 5: Deployment → Day 2

GROUP MEMBERS



PRINCE

TEAM LEADER : FRONTEND AND DASHBOARD



AMAN

ROLE : DATA BASE AND HOSTING



PARTH

ROLE: BACKEND AND API



ADITIYA

ROLE :DATA PROCESSING

CONCLUSION

- Protecting coasts, saving lives, and Preserving nature
- Through AI-powered early alerts.”
- Team Synapse'26 | HackOut’25